WHITEPAPER

Decoding the RFID Landscape:

Exploring Ecosystem, Challenges, and Use Cases

Praful Khandelwal
Senior Architect, Sasken Technologies Ltd.

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Abstract

The white paper explores the Radio Frequency Identification (RFID) ecosystem, looking at various scenarios and its wide-ranging applications across different industries. It discusses the challenges you might face with RFID setups and possible ways to mitigate those challenges.

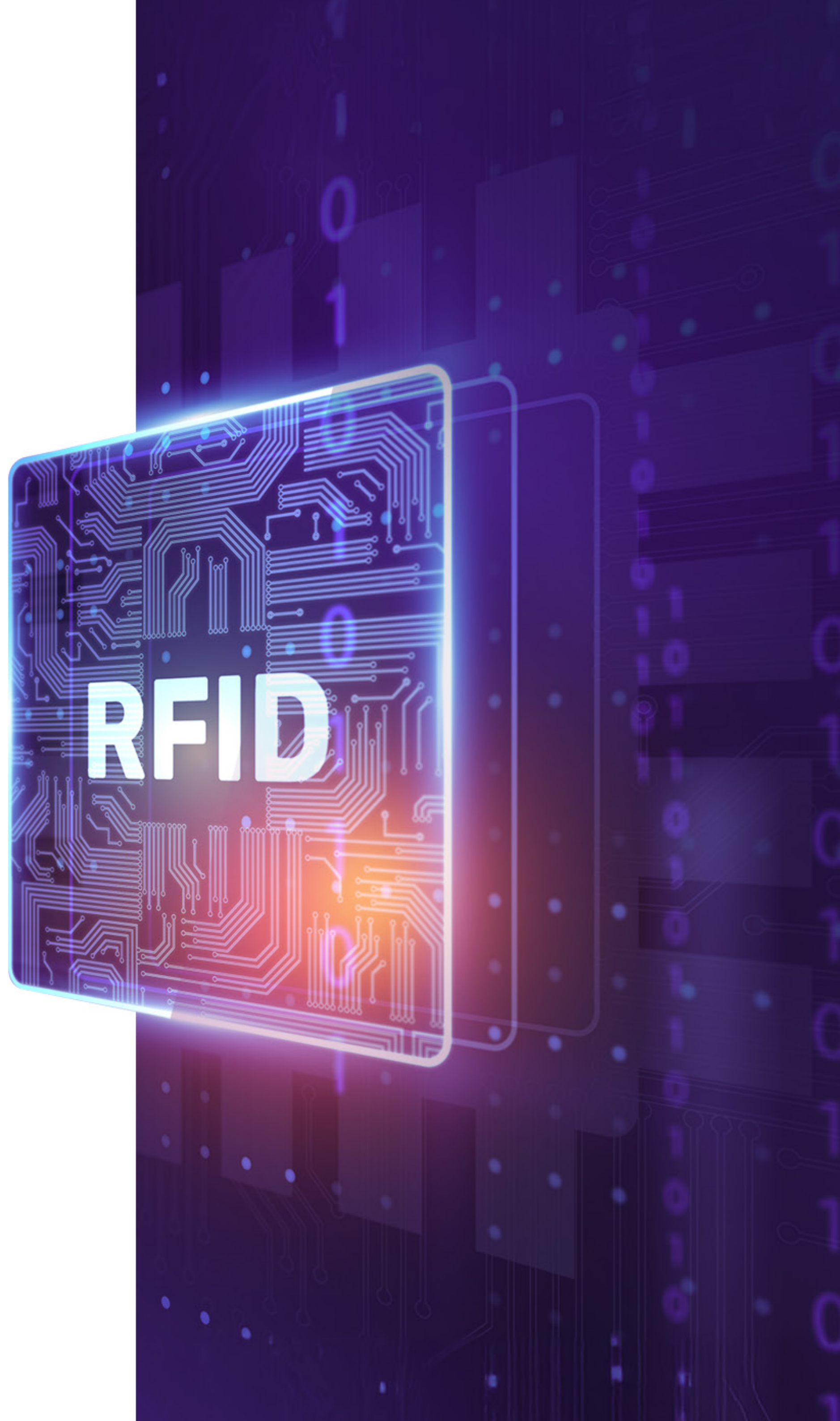
The paper compares RFID with traditional barcode systems and explains how RFID systems integrate with the cloud to provide end-to-end traceability solutions.

Introduction

Let's understand how RFID technology has become a game-changer for automating asset tracking and inventory management in many industries. Unlike traditional barcode systems, RFID lets you capture data from multiple items together over a longer range, without requiring a direct line of sight, making it very efficient.

Imagine a big retail chain managing thousands of products across various stores and warehouses. With barcodes, employees have to scan each item manually—it's slow and error prone. But with RFID, they can scan hundreds of items instantly, simplifying inventory management significantly. This means better control of stock, fewer out-of-stock scenarios, and happier customers.

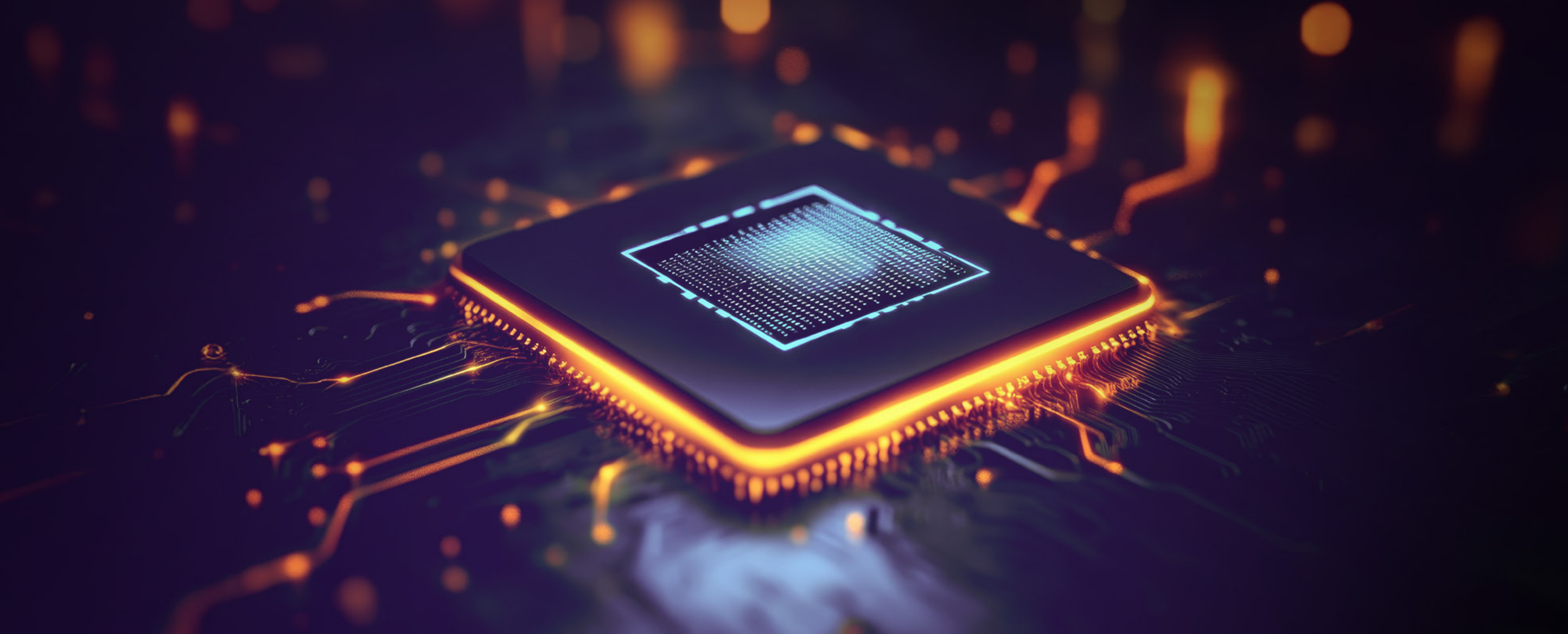
This sets the stage for diving deep into RFID, discussing its benefits and challenges, and strategizing successful rollouts.



Problem Statement

RFID technology sounds like a magic wand for your warehouse, right? Just wave it and know where everything is. But getting it going isn't that easy. Companies face hurdles like hefty upfront costs, integration complexities with existing IT infrastructure, signal interference, and concerns over data security and privacy.

These challenges can really slow things down, stopping businesses from fully adopting RFID to streamline operations and make smarter decisions.



Business Impact

As companies turn towards RFID, here are major impact areas from business standpoint:

- **Operational visibility:**
  RFID gives you real-time tracking and control, essential for staying competitive in today's fast-paced markets.

- **Inventory accuracy:**
  RFID boosts accuracy, ensuring you're always on top of stock levels without the guesswork. This keeps customers happy and sales flowing.

- **Operational efficiency:**
  Automating with RFID cuts down on manual tasks, saving time and resources for more strategic business moves.

- **Regulatory compliance:**
  Stay ahead of regulations with RFID's robust data security measures, building trust and reliability with customers.

- **Strategic advantage:**
  Embracing RFID streamlines your supply chain, improves service speed, and positions you as a leader in your industry.

Understanding the RFID Ecosystem

The RFID ecosystem comprises several key components that work together to enable the identification, tracking, and management of products or assets using RFID technology.

RFID Tag

- Tags are attached to or embedded in products or assets to uniquely identify them
- They can be passive (powered by the reader's signal), active (have their own power source), or semi-passive

Reader Antenna

- An antenna connected to the RFID reader emits radio waves and receives RF (radio frequency) signals from RFID tags
- Antennas come in various forms (e.g., linear, circular) depending on the application and environment

RFID Reader

- The device that communicates with RFID tags via radio waves
- Readers transmit signals to activate tags and receive data back from them
- They can be fixed (stationary) or handheld, depending on the deployment needs

Edge Gateway

- Acts as an intermediary between RFID readers and the cloud software
- Collects data from multiple readers over HTTP, MQTT or TCP/IP and preprocesses or filters it before sending it to the cloud
- Provides local processing capabilities to reduce latency and bandwidth usage

Cloud Software

- Centralized software platform that manages and analyzes data collected from RFID systems
- Stores tag information, tracks movements, and provides real-time visibility and analytics
- Enables integration with other enterprise systems for broader business intelligence and automation
- Can directly receive the data from the reader or via the edge gateway over HTTPS or MQTT

RFID Tags

RFID tags are categorized into Low Frequency (LF), High Frequency (HF), and Ultra-High Frequency (UHF) tags, each suited for different applications based on their frequency range and operational characteristics. Active tags have their own power source for longer-range communication, while passive tags rely on energy from RFID readers. Near Field Communication (NFC) tags, a subset of HF RFID, enable short-range interactions and are commonly used in contactless payments and mobile devices. Choosing the right type of RFID tag depends on factors such as required read range, environmental conditions, and specific application needs.

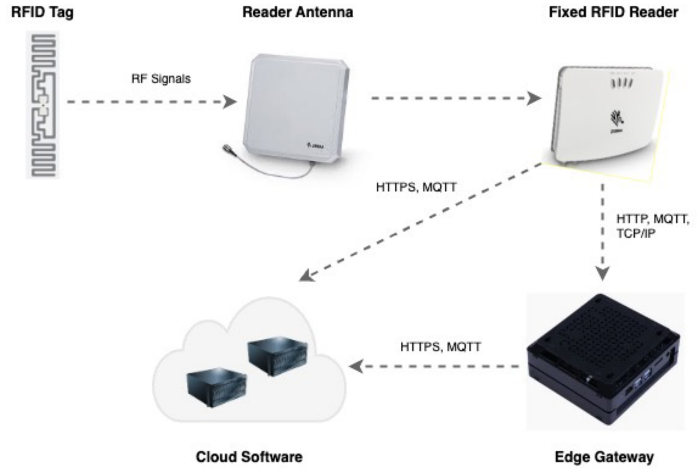
RFID Readers

RFID readers come in various types tailored for specific applications: fixed readers for stationary installations like warehouse entrances, handheld readers for mobile operations, integrated readers embedded in devices, USB readers for direct computer connections, Bluetooth readers for wireless mobility, and differentiating between fixed-mount and handheld options based on mobility and installation needs. Choosing the right RFID reader type hinges on factors such as mobility requirements, operational environments, and integration capabilities with existing systems.

Fixed RFID Reader System Data Flow

Here’s representing the RFID system comprising above components with fixed RFID reader.

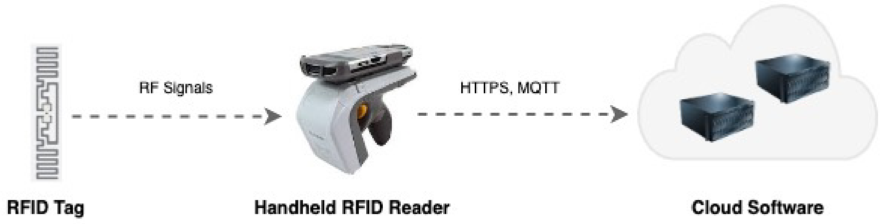
1. RFID tag transmits data to fixed RFID reader in the form of RF signals through the antenna attached to the reader
2. RFID reader decodes the signal, interprets the tag’s data, and processes it before transmitting it either to an edge gateway or directly to the cloud, depending on the setup. Most modern fixed readers can communicate with software systems over HTTP, MQTT, and TCP/IP protocols
3. Edge gateway handles local processing before transmitting data to the cloud. This can be done by either writing custom modules for data processing and then sending it over HTTPS or MQTT to the cloud, or by leveraging IoT edge services such as Azure IoT Edge or Amazon IoT Greengrass
4. Finally, the cloud software processes the data and generates insights



In addition to the core components of RFID systems like tags, antennas, readers, edge gateways, and cloud software, peripheral devices such as light stacks can be attached to fixed RFID readers to enhance functionality and provide visual indicators.

Handheld RFID Reader System Data Flow

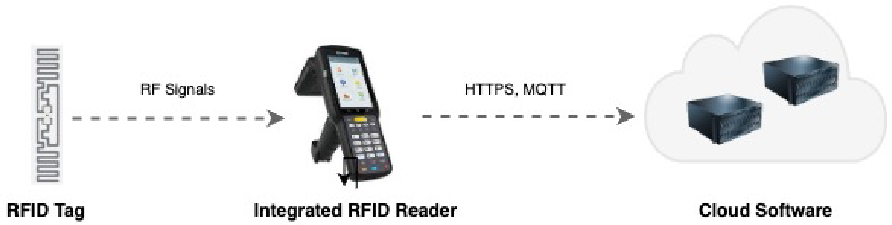
Next, we explore a variation of the RFID system that uses a handheld reader instead of a fixed reader.



1. RFID tag transmits data to handheld RFID reader in the form of RF signals through the antenna integrated with the reader
2. RFID reader decodes the signal, interprets the tag's data, and processes it before transmitting it to paired mobile device over Bluetooth
3. In this setup, the mobile device functions as a gateway, handling local processing before transmitting data to the cloud. A custom mobile app can be developed to run on the device, integrating with the reader through its SDK to retrieve the data. Mobile app can send data over HTTPS or MQTT to the cloud for further processing
4. Finally, the cloud software processes the data and generates insights

Integrated RFID Reader System Data Flow

Finally, we discuss a different configuration that involves an integrated RFID reader.

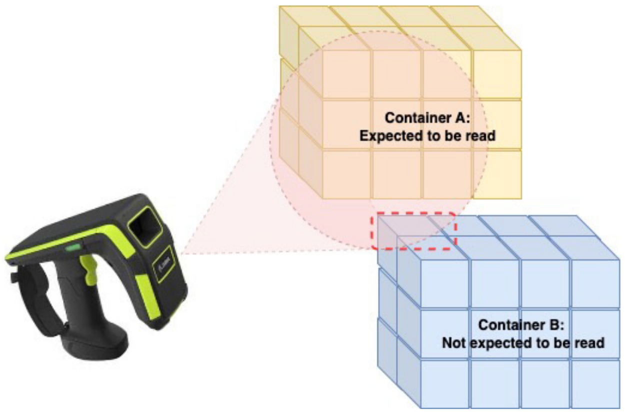


1. RFID tag transmits data to integrated RFID reader in the form of RF signals through the built-in antenna
2. Integrated RFID reader decodes the signal, interprets the tag's data, and processes it internally. The reader can run custom mobile apps to handle local processing of the data
3. Mobile app can send data over HTTPS or MQTT to the cloud for further processing
4. Finally, the cloud software processes the data and generates insights

RFID Challenges and Mitigation Strategies

Over Scans

Due to the extended read range of RFID readers and their ability to scan tags without requiring direct line of sight, there is a possibility of unintentionally scanning additional tags. RFID readers can pick up signals from any tags within their vicinity. This means that tags not intended for the current scan operation might be read, leading to data inaccuracies and potential operational issues.

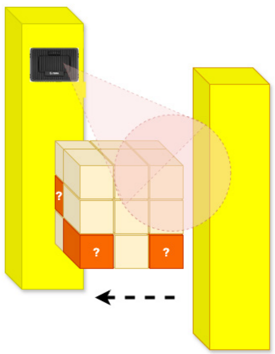


For example, in a warehouse setting, the RFID reader might pick up tags from items stored on adjacent shelves or in nearby containers, resulting in a mix-up of inventory data. This can complicate inventory management, tracking, and auditing processes, necessitating additional filtering and validation steps to ensure data integrity.

To mitigate this issue, strategic placement of RFID readers and careful planning of the scanning environment are crucial. Utilizing RFID shielding, optimizing reader power settings, and implementing software filters can help control and limit the read range to target only the desired tags.

Under Scans

In RFID systems, under scanning can occur when tags are not detected during a scan operation, resulting in incomplete data capture. This issue is often caused by short scan times during high-speed operations or lower read ranges of RFID readers, leading to missed tags that are either too far from the reader or not exposed long enough to be read.

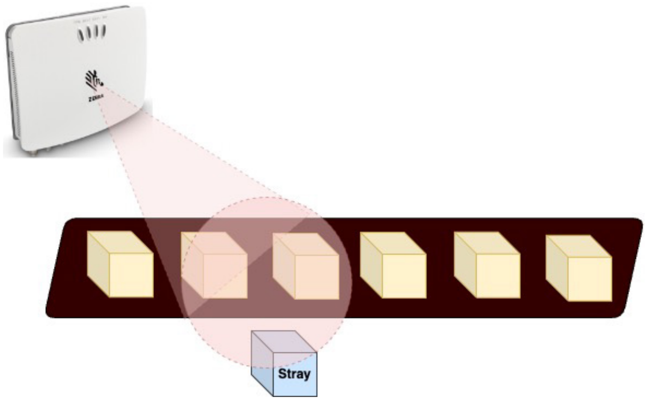


For example, consider a scenario where a pallet loaded with many items is driven quickly through a dock portal equipped with a fixed RFID reader. Because of the high speed and the density of tags, the reader might miss some tags, leading to incomplete scans of the pallet's contents.

There are a few different ways to mitigate this issue, which includes increasing the dwell time by slowing down, extending reading zone by placing more readers and antennas, adjusting reader's sensitivity and power settings, positioning the readers and antennas appropriately to maximize the readability, etc. Most optimal techniques should be adopted to keep the infrastructure cost under check (for example, installing more readers would increase the cost).

Stray Reads

Stray reads occur when RFID readers inadvertently scan tags that are not intended to be read. This can happen when tags that should have been disposed of, removed, or moved to a different location are still within the range of the RFID reader.

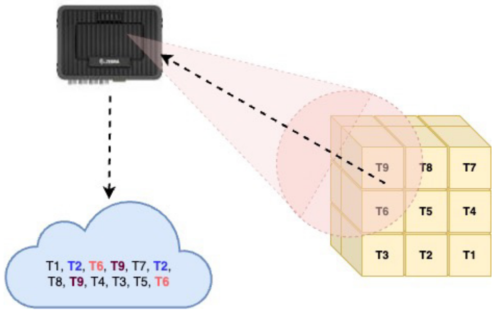


For example, in a conveyor system, boxes with RFID tags are transported along the conveyor belt. Occasionally, a box might fall off the conveyor and remain below it. The fixed RFID reader installed above the conveyor continues to pick up the tag from the stray box, leading to incorrect updates as the system registers the stray box as still being on the conveyor. It may also happen due to someone accidentally placing some other item with RFID tag near to the conveyer.

Again, there are a bunch of techniques that can be employed to mitigate the above situation. These include regularly inspecting and clearing any stray items, adjusting reader range to keep focus on top of conveyer as much as possible, installing physical barriers around conveyer, and applying software filters to detect and ignore stray reads.

Repeated Scans

Repeated scans occur when the same RFID tag is read multiple times by the reader, leading to duplicate entries in the system. This can happen when items are stationary within the reading zone or when they pass through the same reader multiple times.

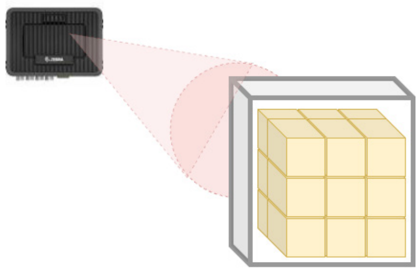


Let's say in a staging area, pallets with RFID-tagged boxes are placed near an RFID reader while awaiting further processing or loading. If the reader is on and continually scans the stationary tags, it can result in multiple reads of the same tag, creating duplicate entries in the system.

Ways to tackle the above situation involve configuring the readers with minimum intervals between scans and using software to filter out duplicates based on timestamps and tag IDs. Adjusting reader placement and antenna orientation to focus on entry and exit and implementing operational procedures to avoid prolonged stationary periods within the reader's range will also help.

Physical Obstructions

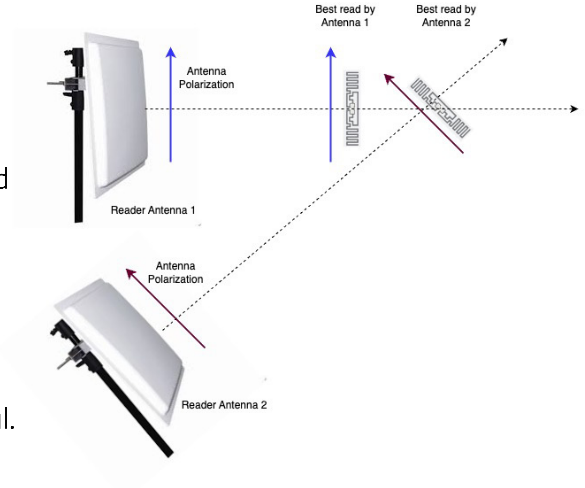
Scan failures can occur when RFID signals are obstructed by materials or objects between the RFID tag and the reader, leading to unsuccessful reads. This can be caused by various mediums such as liquids, metals, or dense materials that interfere with the radio frequency signals, resulting in reflection, scattering or absorption, and not letting the signals reach the tags.



To mitigate this, place tags in optimal positions or use dual tagging to counter obstruction, adjust reader and antenna placement for better coverage, and use materials and technologies designed to minimize signal interference. Employ RFID-enabled or active tags to overcome challenging environments.

Tag Orientation

The orientation of an RFID tag relative to the reader's antenna can significantly impact the read range and success rate due to the polarization of the RFID signal. RFID antennas emit electromagnetic fields that have a specific polarization, either linear or circular. If an RFID tag is not aligned with the polarization of the reader's antenna, such as being perpendicular instead of parallel to the electric field lines, the signal strength can weaken. This misalignment leads to a mismatch in the electromagnetic coupling, resulting in failed reads or reduced read accuracy.



Mitigation involves standardizing tag placement to ensure consistent orientation, using multiple or adjustable antennas to cover various angles, and opting for omnidirectional or dual-antenna tags to improve read success. Conducting regular audits to maintain proper tag orientation will also be helpful.

Signal Interference

RFID systems can experience interference from other devices that operate on similar radio frequencies, such as Wi-Fi routers, Bluetooth devices, or other RFID systems. This interference can cause signal degradation, reducing the accuracy and reliability of RFID reads.
In a busy warehouse, Wi-Fi networks and Bluetooth-enabled equipment operate on the same frequency as the RFID system. The overlapping frequencies cause signal interference, resulting in missed or incorrect reads when tracking items with RFID tags.

Using dedicated frequencies or dynamic channel selection to avoid interference, physically separating RFID systems from other RF devices, optimizing antenna placement, and employing interference filters to ensure reliable RFID operations would be the way to go.

Damaged Tags

RFID tags can get damaged due to physical wear and tear, environmental conditions, or mishandling during operation. Damaged tags may have broken antennas, cracked casings, or other defects that prevent them from getting read successfully or transmitting data properly. In a manufacturing plant or at construction site, RFID tags attached to heavy machinery may get damaged due to exposure to heat, moisture, or impact during operations. A damaged tag may fail to transmit identification data, leading to errors in tracking the equipment.

For such scenarios, durable RFID tags should be considered that are suitable for the operational environment, protective enclosures should be used to shield tags from physical damage, and regular inspection and replacement of damaged tags should be carried out.

Faulty Reader

A faulty RFID reader can disrupt operations by failing to accurately read RFID tags. Issues such as malfunctioning antennas, hardware failures, or communication errors can lead to inconsistent or incomplete data capture. For example, in a logistics warehouse, a faulty RFID reader intermittently fails to detect tags on incoming shipments, resulting in delays and errors in inventory management. The malfunction may be caused due to antenna misalignment or electronic component failure.

It requires standard procedures to tackle such things. Conduct regular maintenance to detect issues proactively, maintain backup RFID systems for continuity, monitor and troubleshoot issues if they occur, and collaborate with vendors for efficient resolution of technical faults.

Tag Collisions

In environments with high tag density, RFID signals from multiple tags can collide or interfere with each other. This collision can lead to signal overlap, resulting in read errors or incomplete data capture.

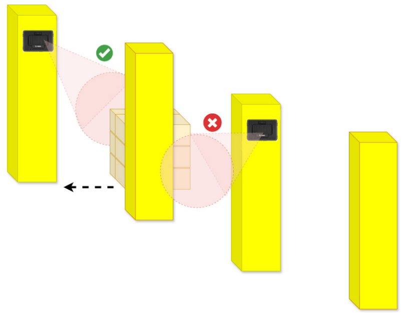
For instance, consider a large-scale event where attendees wear RFID-enabled badges. Multiple tags may be within range of a single reader simultaneously and signal collisions can occur as the reader attempts to process signals from numerous tags simultaneously, potentially leading to missed reads or data inaccuracies.

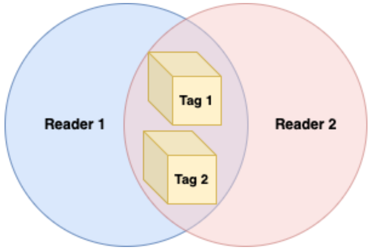
This may require using anti-collision algorithms and optimizing reader configurations to manage simultaneous tag signals effectively. Position antennas strategically and employ sequential tag reading techniques to minimize collisions and ensure accurate RFID data capture in high-density environments.

Unauthorized Reads

Unauthorized reads occur when RFID tags intended for one reader are inadvertently detected by nearby readers operating on the same frequency. This can lead to data duplication or incorrect tracking if tags are scanned by readers not assigned to them.

For example, in a facility with multiple dock doors equipped with RFID readers, tags from products being loaded at one dock door may be unintentionally scanned by adjacent dock door readers. This unauthorized reading can lead to inaccuracies.





This can also be construed as a reader collision problem where more than one reader is trying to read the same set of tags whereas only one of them is authorized to do so.

To mitigate this, physically isolate readers or adjust configurations to minimize overlap, implement frequency management techniques to ensure unique operational channels, and employ authentication and authorization to secure RFID communications and prevent unauthorized reads effectively.

Glimpse into Common RFID Use Cases

Let's take a quick look at some practical use cases which are based on RFID technology:

Inventory Management:
RFID keeps track of stock in stores and warehouses, ensuring shelves stay stocked and inventory is accurate

Asset Tracking:
Industries like construction and logistics use RFID to monitor tools, equipment, and even vehicles, improving efficiency and preventing loss

Access Control:
RFID cards or tags provide secure access to buildings and rooms, replacing traditional keys and enhancing security protocols

Logistics and Supply Chain:
From warehouses to shipping containers, RFID tracks goods in real-time, optimizing logistics and reducing delays

Retail:
RFID speeds up checkout processes in retail stores, enhancing customer experience and reducing waiting times

Livestock Management:
Farms use RFID tags to identify and monitor livestock, ensuring traceability and improving breeding and health management

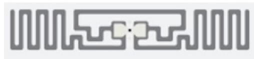
Healthcare Applications:
Hospitals utilize RFID to manage medical equipment, track patient records, and enhance medication administration, improving overall efficiency and patient care

Transportation:
RFID technology simplifies toll collection on highways, enabling faster and more convenient travel without cash or tickets

Examples above are just a glimpse. RFID holds significant potential and can deliver substantial benefits when implemented effectively.
So, are you ready to take a plunge?

RFID or Barcode: Which one is better?

Curious to know how barcodes measure up against RFID? Let's dive right into it!

Feature	Barcode	RFID
Forms / Layout	1D Barcode 2D Barcode – Data Matrix 2D Barcode – QR Code Stacked Barcode	 RFID Inlay  RFID Label (Inlay + Label having barcode and other text) RFID Card  RFID Wristband
Reading Equipment	Barcode reader, mobile camera	RFID reader (handheld or fixed)
Reading Method	Single; Line of sight needed	Multiple; Line of sight not needed
Read Range	Short; Varies based on reader settings and barcode type (1D or 2D)	Longer than barcode; Varies based on reader settings and tag type (active or passive)
Durability	Less durable; Directly exposed as they require line of sight, printed on paper or adhesive labels which can be damaged due to scratches, water, general wear and tear and other environmental factors	More durable; Can be attached inside or embedded as they do not require direct line of sight, generally designed to withstand harsh environmental conditions
Maximum Storage Capacity	Up to 7,089 characters on 2D barcode	128 kb on active tags

Feature	Barcode	RFID
Interference	Low due to single read with line of sight and over short range	Could be higher due to simultaneous reads leading to tag collisions, and presence of mediums such as metals or liquids
Security	More secure; Require line of sight which prevents unauthorized scanning, short read range prevents accidental reads, barcodes do not emit data, limited information stored on barcodes	Potential security concerns; Can be read from distance allowing unauthorized access and unwanted reads, unauthorized readers can skim information from RFID tags within their range, more data available on RFID tags, active tags can transmit data over longer distances
Rewriting	Data remains static	Tags can be rewritten
Overall Complexity	Low	High
Cost	Comparatively lower; barcode scanners are less expensive than RFID readers, with minimal infrastructure set up costs and lower maintenance costs	Comparatively higher; RFID readers are costlier, requiring additional infrastructure such as antennas and middleware, and incur higher maintenance expenses

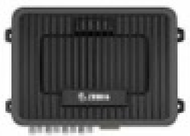
We can use RFID and barcode technologies together to get the best of both worlds. For instance, we might scan barcode labels to get the serial numbers of containers or pallets, and then use an RFID reader to do a wave scan and read all the items beneath them. In fact, many RFID labels are printed with both an RFID inlay and a barcode.

Since it's not always easy to set up RFID everywhere, we sometimes resort to barcodes for tasks such as scanning trucks at gates, processing documents, and other similar scenarios.

How Sasken Stacks Up in the RFID Space

Device Integration Capabilities

Sasken has integrated with multiple different devices in the RFID space using various integration mechanisms and successfully managing the associated complexities.

Device Type	Devices	Integration Details
Fixed RFID Readers	 **Zebra FX9600**  **Zebra FX7500**  **Impinj R700**  **CSL CS463**	• MQTT messages (tag reads) from reader to EMQX broker on edge device or EMQX broker on Azure or Azure IoT Hub • HTTP Post messages (tag reads) over webhook to API endpoints on Azure • Management and Control messages to reader over API or command interface
Handheld RFID Readers	 **Zebra RFD8500**  **Zebra MC3300**  **Zebra TC53/TC58**	• Handheld reader paired with mobile device over bluetooth • SDK based integration between mobile app and reader • Integrated reader with embedded mobile interface to directly run mobile app • Messages (tag reads) from reader to cloud (Azure) via mobile app over HTTP

Device Type	Devices	Integration Details
RFID Printers	 **Zebra ZT410**  **Printronix T4000**	• Generating ZPL files by merging the print templates and scanned/derived information • Printing RFID labels on RFID printers from mobile app by sending ZPL files over TCP/IP socket
Light Stacks	 **Patlite LA6-POE**  **Patlite NHL-3FV2**	• Light stack devices installed in the facility to provide visual indication of success and failure • Controlled from cloud, edge or fixed RFID reader through REST APIs, commands and GPO (General Purpose Output) triggers
Data Loggers	 **Hobo Temperature Logger**	• Data logger used to record temperature readings and pass on those readings to mobile app when paired over bluetooth
IoT Edge Devices	 **Dell OptiPlex**  **Intel NUC**  **Raspberry Pi**	• Edge devices to run EMQX broker or Azure IoT Edge to locally consume tag reads and pre-process them before sending over to cloud

Use Cases Implemented

Here are some critical use cases Sasken has developed leveraging RFID technology, with many surrounding applications.

Use Case	Description
Digitization	Creating digital twins of physical items, identifying them at the class, batch, or instance level. "Class" refers to product code or SKU, "batch" refers to the production lot, and "instance" refers to the unique serial number of each item
Supply Chain Tracking	Tracking items as they move through the supply chain by scanning and recording critical events such as loaded, in-transit, and delivered
Asset Tracking	Monitoring pallets and trailers, including associated parameters such as location and temperature
Inventory Management	Conducting cycle counts to check inventory levels and identify short or out-of-stock items. This can be done manually using handheld RFID readers or automatically with fixed RFID readers
Facility Tracking	Tracking items and pallets within a facility as they move from one area to another, such as from unloading to staging to cooling
Palletization	Shrink-wrapping items together as they are read by fixed RFID readers, and performing validations to ensure items are in good condition regarding weight, expiry, quantity, etc
Trailer Load	Loading items or pallets onto trailers for delivery, read by fixed RFID readers mounted at dock doors, and performing validations to ensure items are in good condition regarding weight, expiry, quantity, etc
Picking	Locating and picking items using handheld RFID readers by monitoring beeps and Received Signal Strength Indicator (RSSI) values
Recall/On-Hold Management	Placing items in recall or on-hold status for any known issues to prevent further movement in the supply chain
Device Management	Managing various devices in RFID setups, including fixed and handheld RFID readers, edge devices, light stacks, barcode readers, printers, mobile devices, and more

Conclusion

In conclusion, RFID technology brings game-changing benefits by transforming how businesses track and manage assets. Its knack for providing real-time, spot-on data without getting too close to items boosts efficiency and decision-making across different industries. Sure, there are hurdles upfront with adoption and integration, but once businesses nail it, RFID pays off big time. Think better inventory checks, smoother operations, and happier customers. As companies push the envelope in supply chain innovation, RFID remains a must-have for staying sharp and ahead in today's fast-paced business world.

Sasken offers extensive expertise not only in developing and integrating RFID-based solutions seamlessly but also in-depth understanding of various RFID setups, scenarios, and challenges.

About The Author

Praful Khandelwal, Senior Architect at Sasken, has more than 20 years of experience in the IT industry across multiple domains and technologies.

Praful is passionate about technology and his expertise lies in architecting and designing complex multi-cloud solutions incorporating Microservices, IoT, Big Data, Web apps, Mobile apps, System integrations, and hardware integrations.

He has hands-on experience across a wide range of tools and technologies including Python, JavaScript, Hadoop, Spark, Scala, Airflow, N8n, SQL/NoSQL/Timeseries databases, Presto, AWS, Azure, Kafka, EMQX, Docker, Kubernetes and others.